Advancing Lung Cancer Diagnostics through Multimodal Deep Learning: Optimal Fusion of 3D Radiological Imaging and Clinical Data with LLM-Driven Interpretability

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ABSTRACT

Background/Objective:
Early lung-cancer detection is limited by variability in CT interpretation and the absence of patient context in image-only AI. PulmoAID aims to improve screening decision support by fusing chest CT with clinical/demographic data (and optional LLM-derived text features) while providing transparent explanations for clinicians and patients.

Methods:
Public, de-identified CT datasets were preprocessed (slice selection, normalization, lung masking). Image features were learned via transfer learning (2D VGG16; 3D CNNs). Tabular clinical/demographic features were engineered and combined with optional LLM-derived embeddings. Unimodal baselines (CT-only; clinical-only) were compared to multimodal fusion using stratified splits. Primary metric: F1-score; secondary: accuracy, precision, recall. Explainability used Grad-CAM (spatial saliency) and SHAP (feature attribution). A lightweight web app delivered calibrated risk estimates and plain-language rationales.

Results:
The multimodal model achieved F1 = 0.88 and accuracy = 91% on a held-out test set, outperforming CT-only (F1 ≈ 0.62) and clinical-only (F1 ≈ 0.70) baselines, with fewer false negatives and false positives. Inference latency was <2 s with preloaded weights. Grad-CAM localized suspicious regions on CT, SHAP identified influential risk factors (e.g., smoking history, age), and the LLM layer summarized case-specific rationale.

Conclusions:
Multimodal fusion with transparent explanations improved performance over unimodal pipelines and may support earlier, more consistent detection across diverse settings. Future work includes multi-site external validation, uncertainty calibration, and PACS/EHR integration.

Keywords:
Lung cancer; Multimodal fusion; CT imaging; Clinical data; Explainable AI; Grad-CAM; SHAP; Large language models; Decision support.

INTRODUCTION

Background and Context

Lung Cancer (LC) remains a significant global health issue with high mortality rates, primarily due to late-stage diagnosis. Lung cancer is caused by the abnormal growth of cells in the lungs, forming masses or tumors that can interfere with normal lung function. In 2020, lung cancer accounted for over 1.8 million fatalities worldwide, making it the leading cause of cancer-related deaths globally ((World Health Organization, Cancer fact sheet: Lung cancer statistics. https://www.who.int (2020).)), ((American Cancer Society, What is lung cancer? https://www.cancer.org (2020).)).

Lung tumors are classified based on the type of cells they originate from; for example, adenocarcinoma arises in glandular cells, while squamous cell carcinoma develops in the cells lining of the airways ((National Cancer Institute, Lung cancer types and stages. https://www.cancer.gov (2020).)), ((Mayo Clinic, Non–small cell lung cancer. https://www.mayoclinic.org (2020).)). Tumors are also categorized by their malignancy: benign tumors typically remain localized, whereas malignant tumors grow aggressively and can spread to other parts of the body, significantly lowering survival rates ((American Cancer Society, What is lung cancer? https://www.cancer.org (2020).)) , ((R. L. Siegel, K. D. Miller, A. Jemal, Cancer statistics, 2020. CA Cancer J. Clin. 70, 7–30 (2020).)). Detecting cancerous cells in their initial stages is crucial because it can significantly improve treatment outcomes, devise personalized treatment plans and improve the likelihood of survival. Treating cancer at an advanced stage is often more expensive due to the need for complex treatments, such as chemotherapy or extensive surgery ((6.

D. Jones, R. Patel, Economic burden of misdiagnoses in lung cancer patients. Lancet Oncol. 21, 1456–1462 (2020). doi:10.1016/S1470-2045(20)30389-7)). If this growth is not controlled early on, it can result in death.

Problem Statement and Rationale

Computed tomography (CT) is a non-invasive imaging technology widely used in LC screening because it can produce high-resolution lung images ((Radiological Society of North America, Computed tomography and its role in lung cancer screening. https://www.rsna.org (2020).)), ((Lung Cancer Research Foundation, Challenges in CT-based lung cancer diagnosis, https://www.lungcancerresearchfoundation.org (2020).)), ((F. O'Sullivan, J. R. O'Neill, CT imaging in early lung cancer detection. Radiology 293, 530–546 (2020). doi:10.1148/radiol.2020200456)). Radiologists manually examine CT scans to identify suspicious nodules or tumors, a process that is both time-intensive and subject to inter-observer variability, leading to potential diagnostic delays and errors—especially in high-volume clinical settings. While CT scans are invaluable for identifying suspicious regions, they may not always distinguish malignant tumors from benign conditions such as infections or scarring, creating diagnostic uncertainty ((Lung Cancer Research Foundation, Challenges in CT-based lung cancer diagnosis, https://www.lungcancerresearchfoundation.org (2020).)), ((A. Smith, Y. Wang, Artificial intelligence in lung cancer imaging: Current trends and challenges. J. Thorac. Oncol. 15, 1413–1427 (2020). doi:10.1016/j.jtho.2020.05.008)), ((S. F. Yang, L. M. Chen, Improving CT scan diagnostics for lung cancer in low-resource settings. Int. J. Cancer 147, 3458–3467 (2020))). Moreover, accuracy depends heavily on access to experienced radiologists, which can be limited in resource-constrained regions. To address these challenges, there is a growing need for automated, AI-driven diagnostic systems that not only analyze imaging data but also leverage complementary patient information.

Recent advances in machine learning (ML) and deep learning (DL) have significantly improved the performance of diagnostic approaches in medical imaging ((A. Smith, Y. Wang, Artificial intelligence in lung cancer imaging: Current trends and challenges. J. Thorac. Oncol. 15, 1413–1427 (2020). doi:10.1016/j.jtho.2020.05.008.)). Convolutional Neural Networks (CNNs) such as VGG16 ((K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition. In: International Conference on Learning Representations (ICLR) Workshop (2015). arXiv:1409.1556. doi:10.48550/arXiv.1409.1556)) , and ResNet ((K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778 (2016).)), have achieved good results in detecting and classifying lung nodules on CT scans. While useful, image-only systems fail to leverage complementary patient information such as age, gender, smoking history, comorbidities, and prior diagnoses factors that can provide more accurate predictions, especially in incipient or ambiguous cases ((S.-C. Huang, A. Pareek, S. Seyyedi, I. Banerjee, M. P. Lungren, Fusion of medical imaging and electronic health records using deep learning: A systematic review and implementation guidelines. npj Digit. Med. 3, 136 (2020). doi:10.1038/s41746-020-00341-z.)).

This potential upside has driven increasing interest in multimodal learning, where models combine features from multiple data sources to improve predictive performance. In lung cancer diagnostics, this typically involves fusing CT imaging with structured clinical data (e.g., demographics, lab results). Studies have shown that such combinations can yield higher diagnostic accuracy compared to imaging alone ((S.-C. Huang, A. Pareek, S. Seyyedi, I. Banerjee, M. P. Lungren, Fusion of medical imaging and electronic health records using deep learning: A systematic review and implementation guidelines. npj Digit. Med. 3, 136 (2020). doi:10.1038/s41746-020-00341-z.)), ((N. Nikolaou, D. Salazar, H. RaviPrakash, M. Gonçalves, R. Mulla, N. Burlutskiy, N. Markuzon, A machine learning approach for multimodal data fusion for survival prediction in cancer patients. npj Precis. Oncol. 9, 128 (2025).)). However, many of these frameworks use late fusion strategies, processing each modality independently before pooling predictions at the classification stage. This limits the degree of cross-modal interactions and may underutilize the full predictive power of the combined data ((N. Nikolaou, D. Salazar, H. RaviPrakash, M. Gonçalves, R. Mulla, N. Burlutskiy, N. Markuzon, A machine learning approach for multimodal data fusion for survival prediction in cancer patients. npj Precis. Oncol. 9, 128 (2025).)).

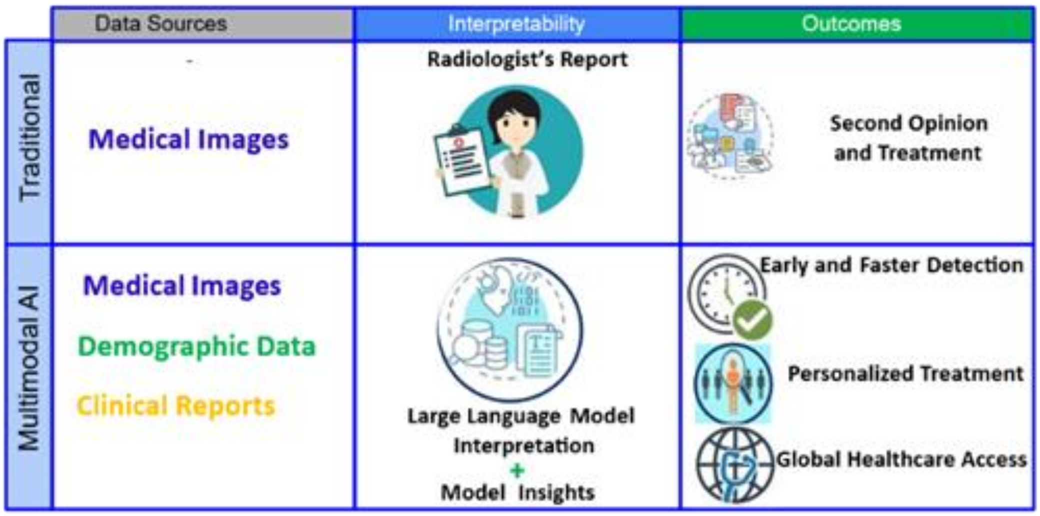

Figure 1: Multi-Modal Approach for Early Detection

Another valuable modality can come from unstructured clinical text, including physician notes and radiology reports. Natural language processing (NLP), and more recently, large language models (LLMs) can extract semantic representations and detect nuanced patterns not explicitly recorded in structured fields. LLMs can identify subtle risk factors, temporal disease patterns, and symptom correlations that are otherwise inaccessible to imaging-based models. Integrating these text-derived features with imaging and structured data offers a richer, more holistic patient representation ((Y. Nam, Multimodal large language models: A new era in medical imaging. Korean J. Radiol. 26, 321–331 (2025). doi:10.3348/kjr.2025.0599)).

Alongside improved prediction, explainability is critical for clinical adoption. Visualization tools such as Gradient-weighted Class Activation Mapping (Grad-CAM) ((R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), 618–626 (2017). doi:10.1109/ICCV.2017.74)) and Shapley Additive exPlanations (SHAP) ((S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems (NeurIPS), Vol. 30 (2017).)) have become standard for interpreting CNN decisions. Yet, these tools often produce technical outputs that are best understood by ML developers, not clinicians. LLMs present an emerging capability to bridge this gap by generating patient-specific, natural language explanations that contextualize predictions within a patient's broader clinical profile ((F. Gaber, M. Shaik, F. Allega, A. J. Bilecz, F. Busch, K. Goon, V. Franke & A. Akalin, Evaluating large language model workflows in clinical decision support. npj Digit. Med. 8, 84 (2025). doi:10.1038/s41746-025-01684-1)).

Multimodal fusion, integrating radiological imaging with structured clinical variables and insights derived from unstructured clinical narratives via large language models (LLMs), offers the potential for more accurate, explainable, and accessible lung cancer diagnostics. This approach captures a broader clinical context, enabling models to detect patterns that may be invisible to unimodal systems, and provides personalized interpretable outputs that support clinician decision-making and help build confidence with patients.

Significance and Purpose

This research advances medical AI for lung-cancer decision support by showing how multimodal fusion of chest CT scans, clinical/demographic data and clinical notes can be useful for better training of the AI models to improve screening accuracy at the patient level. Additionally, this work uses the Large Language Model (LLM) to generate human-readable explanations that highlight the key reasons for why a patient is flagged as high-risk and can help pulmonologists build trust with the patients by promptly starting personalized treatment.

Multimodal Approach for Diagnostics

We designed a multimodal fusion method that combines medical imaging with clinical data and reports to improve the accuracy of early-stage lung cancer detection as described in Figure1. This approach leverages the strengths of each modality to enhance diagnosis and ensure more accurate disease screening. We integrated advanced transfer learning and LLM methodologies to extract insights from rich clinical and imaging data derived from the National Lung Screening Trial (NLST) ((20. National Cancer Institute, National Lung Screening Trial (NLST) dataset. https://www.cancer.gov/types/lung/research/nlst (2011).)).

To develop robust multimodal, LLM-enabled diagnostics, we investigated whether we can efficiently combine radiology (CT scans) with individualized patient data, including demographic, lifestyle, and lab report information, through feature fusion to improve the accuracy and accessibility of early-stage lung cancer diagnostics. We hypothesized that transfer learning models using 3D features extracted from medical images will outperform VGG16-like 2D models applied to individual CT slices in terms of binary classification accuracy. Furthermore, we hypothesized that fusing features extracted from the 3D transfer learning models with features derived from clinical reports via NLP and key demographic predictors will deliver the high fidelity required for personalized medicine.

Objectives

The overall aim of this work is to design a multimodal, LLM-enabled diagnostic framework that fuses radiological, structured clinical, and unstructured text data to deliver accurate, interpretable, and accessible lung cancer diagnostics. This system to be effective in both high-resource and low-resource healthcare environments, where early, explainable screening can reduce mortality and economic burden.

Research focus is to develop and implement a novel, multimodal algorithm for robust and early detection of lung nodules/tumors which outperform existing methods. Key objectives of this study are:

- Devise a higher-fidelity hybrid model using features from CT scans, clinical data and doctor notes.
- Establish LLM-based explainability for improved understanding of the diagnosis with the CT scans based on the personalized clinical data.
- Build a secure, web-based clinical application.

This research introduces a multimodal approach for lung-cancer decision support that fuses chest CT imaging with clinical/demographic data and pairs it with explainable outputs. By leveraging complementary signals across modalities, the method is designed to improve F1-score, reduce false negatives, and increase robustness to real-world variability. The best-performing model is implemented in a secure, interactive web app delivering point-of-care risk estimates and plain-language rationales, enabling clinicians and patients to understand what the model sees and why it recommends a given diagnosis.

Key research questions addressed in this work are (1) What is the most effective method of combining CT scans and patient data based on feature fusion to improve the accuracy and interpretability of early-stage lung cancer diagnostics? (2) What impact do algorithm parameters (e.g., learning rate, dropout fraction) and training cycles have on classification F1 score?

Towards the goal of developing an multimodal lung cancer diagnostic system, current study hypothesizes: If the training dataset is consistent, balanced, and sufficiently diverse, then combining features extracted by transfer-learning models, clinical reports summarized by large language models, and key demographic predictors will yield more accurate diagnostic outcomes than CT scans alone, while also providing valuable insights for precision medicine.

Scope and Limitations

We trained/evaluated models on public, de-identified datasets that may not fully represent the diversity of scanners, acquisition protocols, and patient populations seen in clinical practice. Although we used stratified splits, external multi-site prospective validation has not yet been conducted. The LLM layer can introduce hallucinations or overconfident language; we mitigate with guardrails and templated rationales, but clinician oversight remains essential. PulmoAID is not FDA-cleared and should not be used as an autonomous diagnostic.

Methodology Overview

A multimodal, two-stage AI pipeline was implemented for PulmoAID. First, chest CT studies were preprocessed (slice selection, normalization, lung masking) and encoded with transfer-learned CNNs (2D VGG16; exploratory 3D backbones) to derive imaging features. In the second stage, these features were fused with structured clinical/demographic variables (and optional LLM-derived text embeddings) in a calibrated classifier. Training used stratified splits and an optimized learning rate to ensure stable convergence. The final system was evaluated on a held-out test set (reporting F1, accuracy, precision/recall) and deployed as a lightweight web application (PulmoAID) to demonstrate practical, point-of-care use with explainable outputs (Grad-CAM, SHAP).

RESULTS

We evaluated PulmoAID on a balanced dataset of annotated CT scans and corresponding clinical records, totaling 495 patient samples from the testing partition of the dataset. We assessed model performance using accuracy, precision, recall, and F1 score.

The best model trained on primary data set of 689 patients with 20% validation partition achieved an accuracy of 91%, precision of 86%, recall of 91%, and an F1 score of 88%. The multimodal fusion model outperformed all single-modality baselines: image-only models reached an F1 score of 62%, while structured-data-only models reached 70%. Adding LLM-derived features improved all metrics by 7–10% compared to bimodal models using only imaging and structured data.

Binary Classifier Comparisons

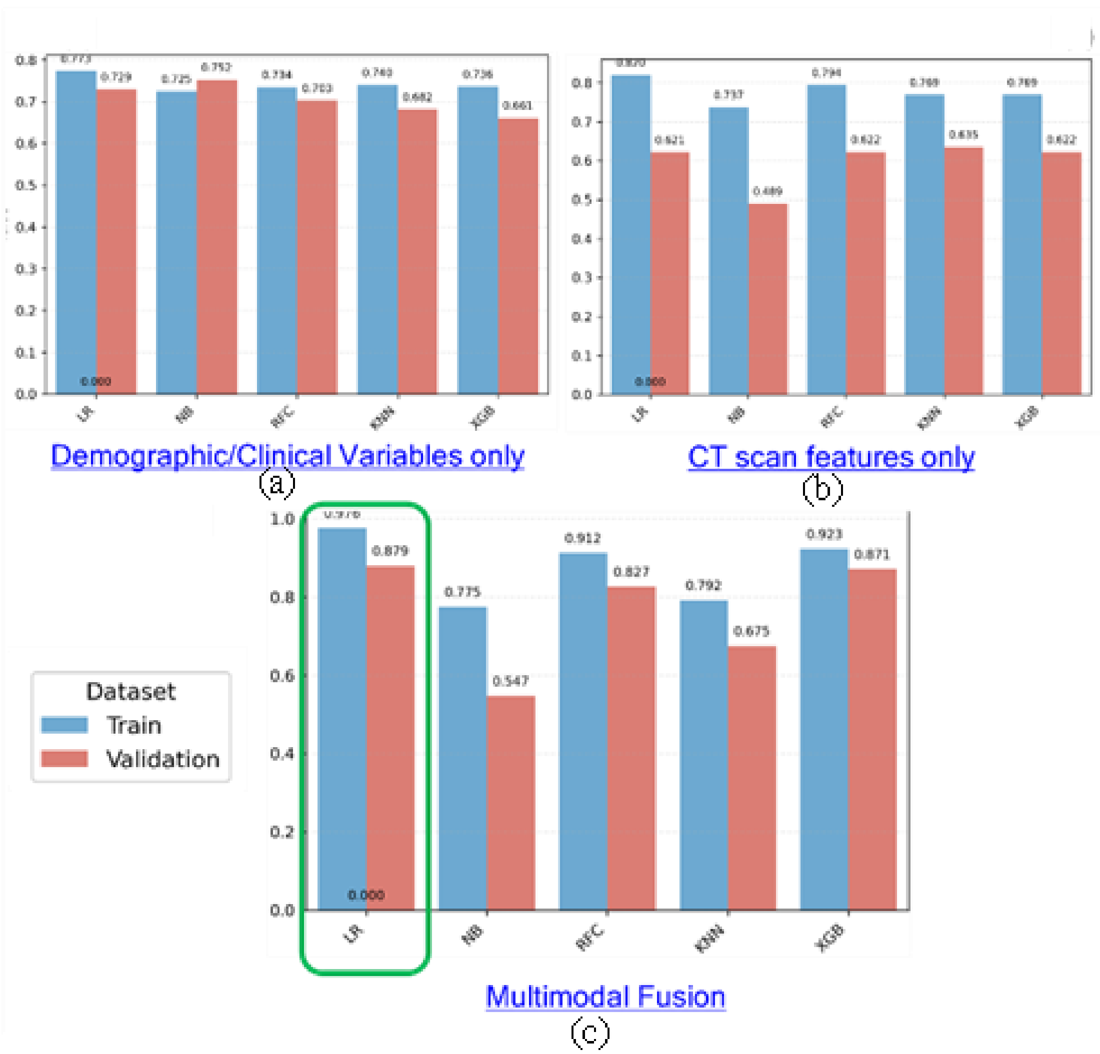

Figure 2: Binary classifier performance (F1 score) for five models: Logistic Regression (LR), Naïve Bayes (NB), Random Forest Classifier (RFC), K-nearest Neighbors (KNN), and XGBoost (XGB), evaluated on three input configurations: (a) demographic/clinical variables only, (b) CT scan features only, and (c) multimodal fusion (clinical + imaging + LLM-derived features). Results are shown for both training and validation datasets.

We compared five binary classifiers—Logistic Regression (LR), Naïve Bayes (NB), Random Forest Classifier (RFC), K-nearest Neighbors (KNN), and XGBoost (XGB)—across three experimental setups:

- Structured demographic/clinical variables only
- CT scan features only
- Multimodal fusion of structured, imaging, and LLM-derived features

For structured variables alone, F1 scores on the testing set ranged from 0.734 (NB) to 0.781 (XGB), with LR achieving 0.752. For CT features alone, performance was lower overall, with NB scoring 0.489 and RFC scoring 0.704 on validation.

The multimodal fusion model achieved the highest scores in every classifier. LR combined with VGG16 features gave the best performance with an F1 score of 0.879 with model accuracy of 91%. RFC and XGB also performed strongly, achieving validation F1 scores of 0.827 and 0.871, respectively. NB showed substantial improvement in the multimodal setting, rising from 0.489 (CT only) to 0.775. These results confirm that multimodal integration consistently boosts classification performance compared to individual modalities as summarized in Figure 2.

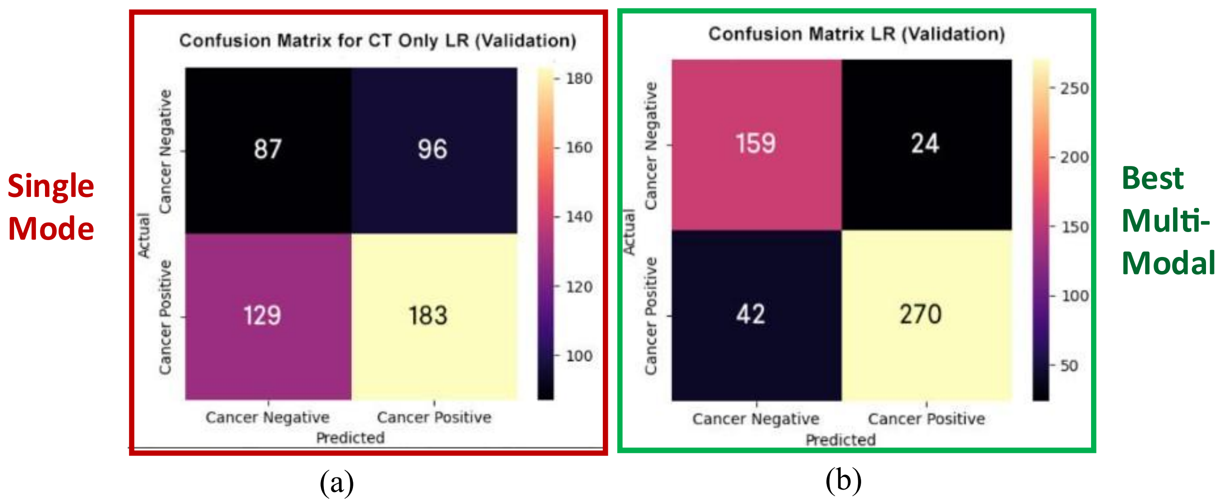

Figure 3: Confusion matrices comparing (a) single-modality (CT-only) and (b) multimodal logistic regression models on the testing set.

In Figure 3, the left confusion matrix represents a logistic regression classifier trained solely on CT-derived radiomic features, exhibiting limited discriminative capacity with high false negative (FN = 129) and false positive (FP = 96) rates on the held-out testing data set of 495 patients. In contrast, the right confusion matrix demonstrates the performance of the optimal multimodal fusion model, which integrates both imaging and structured clinical data. This fusion significantly enhances diagnostic accuracy, reducing false negatives to 42 and false positives to 24, while increasing true positives (TP = 270) and true negatives (TN = 159). The performance improvement underscores the benefit of complementary information from non-imaging clinical features in improving model generalizability and sensitivity to early-stage malignancies.

Interpretability Graphs

Grad-CAM visualizations for original CT scan and 8 extracted features are shown in Figure 4. This highlights the spatial importance of specific image regions that contribute to PulmoAID's prediction, with each heatmap corresponding to a latent feature used in the classification process. Notably, several features exhibit high activation (hotspots) around radiologically significant regions, such as spiculated tumor margins, central lobar masses, and cavitary lesions, confirming the model's ability to focus on clinically meaningful patterns.

As shown in Figure 4, not all features are equally activated for a given case. Some appear dormant, showing minimal heatmap intensity, suggesting they contribute minimally to the classification of that specific subject. This dynamic feature engagement aligns with PulmoAID's personalized recommendations, where the system selectively utilizes the most informative feature representations for each patient. Overall, the visualization supports the interpretability and clinical relevance of the model's internal representations.

SHAP analysis further revealed the relative contributions of non-imaging features. Smoking history, age, and genetic predisposition were among the most influential structured variables, while sentiment indicators from physician notes were top contributors among LLM-derived features.

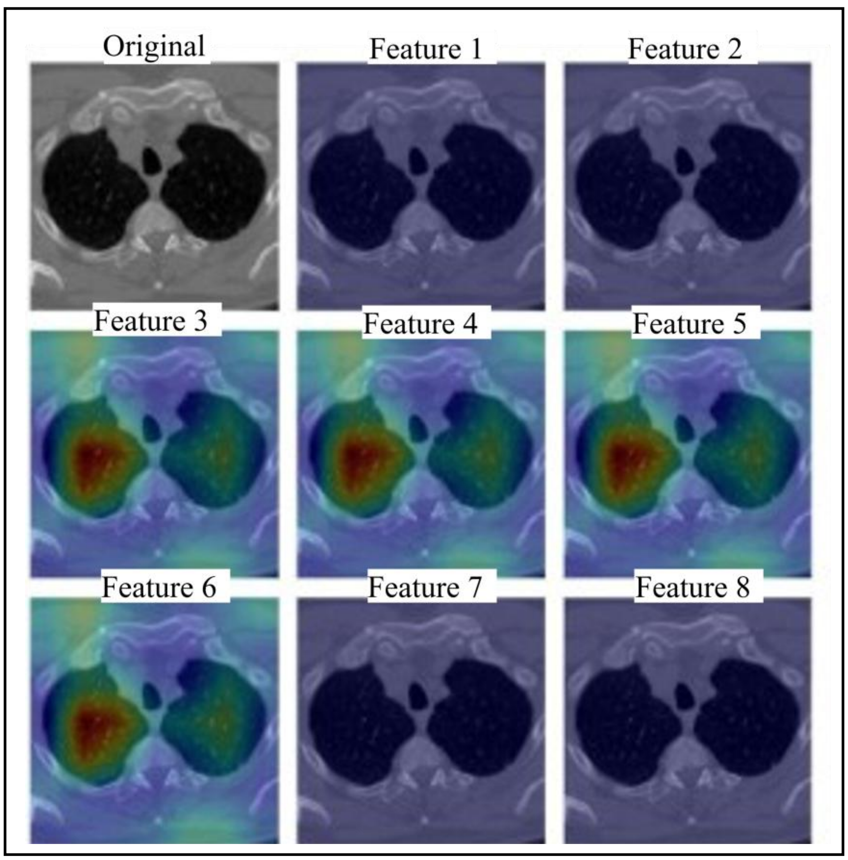

Figure 4: Grad-CAM Visualization of Lung CT Scan Features Contributing to Cancer Prediction. The original CT image (top-left) is overlaid with Grad-CAM heatmaps corresponding to different learned features by the convolutional neural network. Warmer colors (red/yellow) indicate regions with higher importance for model prediction. Features 2 through 6 show strong activations in suspicious pulmonary regions, demonstrating localized attention consistent with radiological assessment.

LLM-enabled Virtual Doctor

To further enable interpretability, PulmoAID features a “Virtual Doctor” conversational engine that takes the inputs and model results to provide a simple explanation of the diagnostic summary and medical insights that patients can understand as depicted in Figure 5. This GenAI-powered feature enhances interpretability and transparency in lung cancer diagnostics by translating complex data into structured, explainable reasoning.

PulmoAID’s deployment as a web application also underwent functional testing with mock user inputs in low-resource scenarios. Response times averaged below 2 seconds for inference, and system robustness held across varied input sizes. This highlights its scalability for global implementation in resource-limited clinical settings.

Overall, these results underscore the effectiveness and reliability of PulmoAID as a clinically relevant tool for enhancing early lung cancer detection through interpretable multimodal AI.

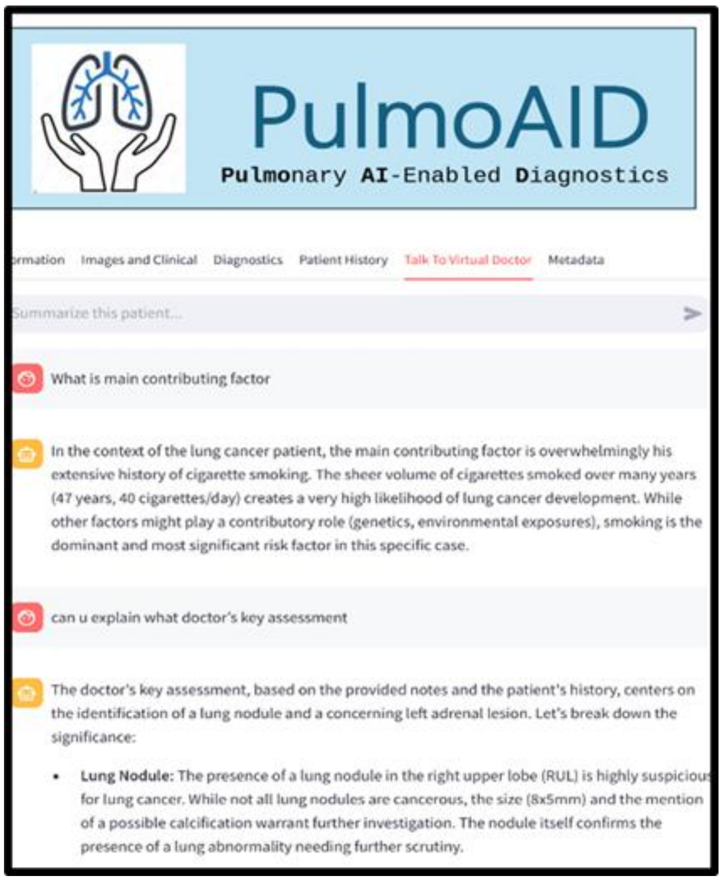

Figure 5: PulmoAID's Virtual Doctor Interface provides AI-driven clinical decision support by summarizing key diagnostic insights from both patient history and imaging data.

System Design and Development

PulmoAID is designed as a modular, end-to-end diagnostic decision support system, delivered via a lightweight web interface to ensure accessibility in both high- and low-resource environments. At its core, the platform follows microservices-based architecture, enabling separation of responsibilities across critical pipelines: data ingestion, preprocessing, inference, visualization, and user interaction as shown in Figure 6. This design facilitates independent development, scalability, and rapid updates of specific components without affecting the entire system. Development and integration were performed using Visual Studio Code, with ETL pipelines implemented in Pandas, deep learning models built in PyTorch, and result visualization handled via Matplotlib. Version control was maintained through GitHub, ensuring reproducibility and collaborative development. All model artifacts and outputs are systematically tracked.

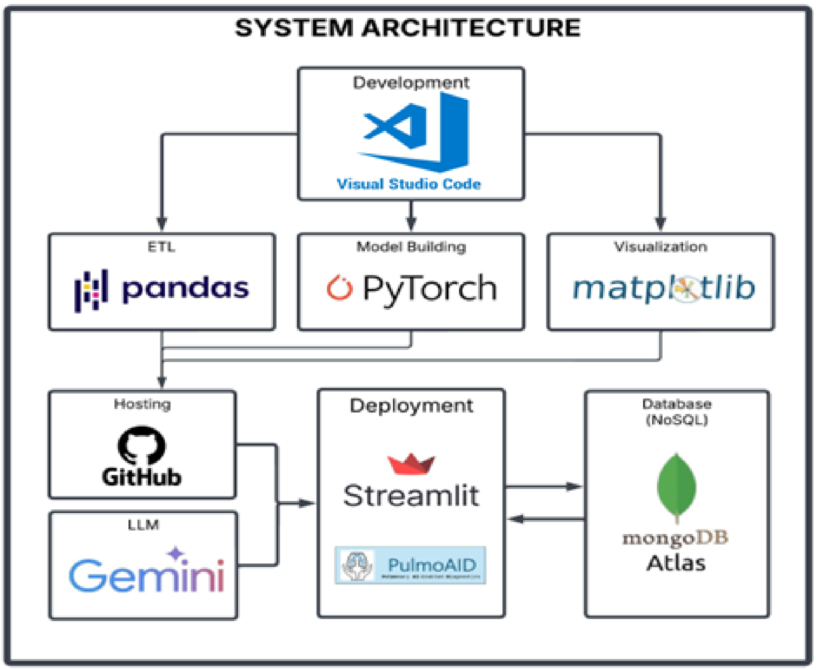

Figure 6: Overview of the end-to-end system architecture for the multimodal lung cancer prediction platform. Visual Studio Code was used for development and integration of data processing (ETL with Pandas), model building (PyTorch), and result visualization (Matplotlib). Model source code is version-controlled and hosted on GitHub. Deployment to end-users is handled via Streamlit and the PulmoAID web application. The system incorporates Gemini as the LLM for generating sentiment features and interacts with a NoSQL MongoDB Atlas database for storing processed data, prediction.

The front-end, developed in Streamlit, supports interactive data input, real-time inference outputs, and multimodal feedback. Users can upload CT scans and structured clinical data, receive diagnostic predictions, and view both Grad-CAM visualizations for explainability and LLM-generated narrative reports using Gemini. These sessions are securely tracked and allow downloadable PDF reports for offline use. Backend services interact with a MongoDB Atlas NoSQL database to store user input, predictions, and logs. The entire system is containerized using Docker, supporting both cloud-native deployment and on-premises installation on hospital servers or research clusters. Compliance with HIPAA-aligned practices is ensured through encryption, access control, and audit trails for each session. For field-readiness, PulmoAID was validated in simulated low-bandwidth environments, demonstrating sub-3-second latency by preloading model weights and compressing outputs. These features position PulmoAID as a scalable, secure, and clinically usable AI diagnostic assistant as shown in Figure 7.

Data Privacy and Safety Considerations

The PulmoAID web application follows secure access control, data encryption, and compliance with data protection standards. It does not allow download or sharing of user-uploaded images. All classification is performed in-session and images are processed temporarily for prediction. No personally identifiable information (PII) is collected or retained.

DISCUSSION

PulmoAID demonstrates the transformative potential of multimodal AI systems in enhancing early lung cancer diagnosis, particularly through the fusion of CT imaging, structured health records, and LLM-derived clinical features. This fusion represents a novel modeling direction in diagnostic AI, balancing technical performance with interpretability and deployment ease. The model's strong predictive performance across diverse metrics, combined with its interpretability features, positions it as a promising clinical tool for augmenting decision-making.

A key strength of PulmoAID lies in its incorporation of contextual information often ignored in traditional radiological AI models. By integrating structured data and LLM-extracted features, the system captures a holistic representation of a patient's health status—bridging the gap between purely visual cues and clinical reasoning as demonstrated in Figure 7. This approach not only improves diagnostic accuracy but also supports explainability in a format familiar to healthcare professionals.

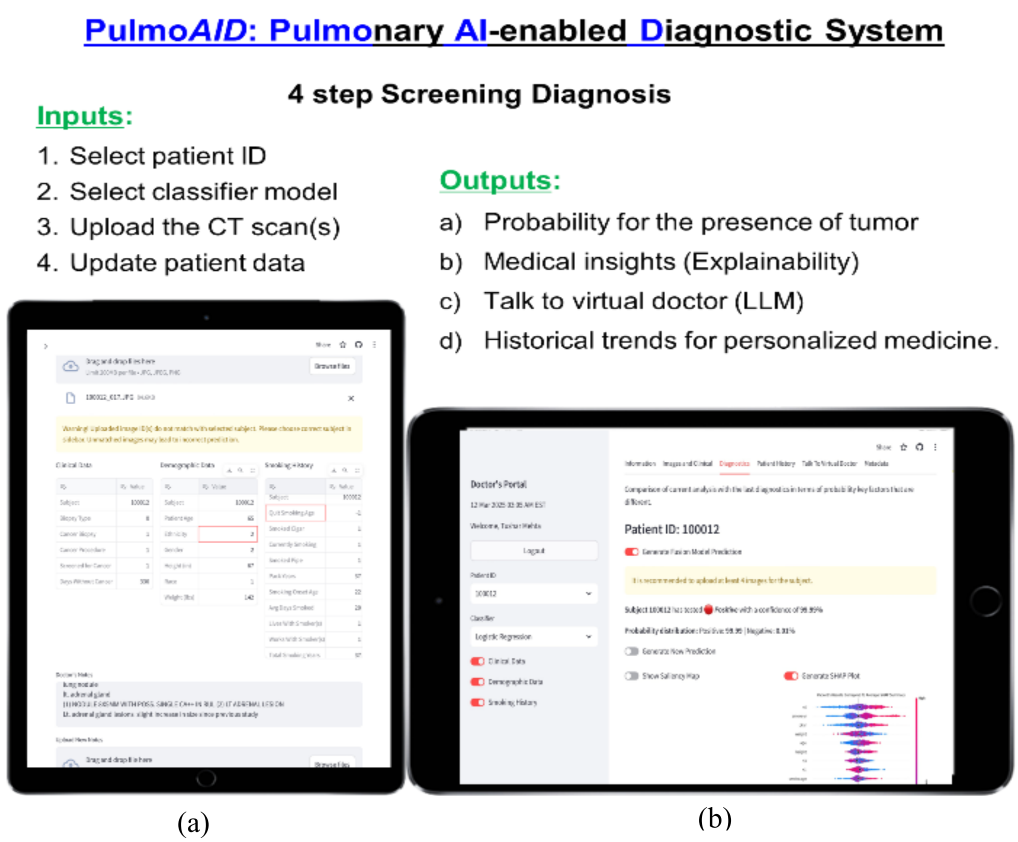

Figure 7: PulmoAID platform interface illustrates the end-to-end flow from data input to model-assisted diagnostic output. (a) Structured patient metadata (e.g., demographics, smoking history) and imaging files are uploaded through a secure web interface. (b) Outputs include risk prediction, Grad-CAM visualizations for interpretability, and an LLM-generated natural explanation, forming a multimodal, explainable diagnostic report designed for clinician review and decision support.

Moreover, the system's web-based deployment and modular architecture enable wide accessibility and ease of integration. With its compatibility across platforms and support for mobile usage, PulmoAID can be deployed in under-resourced regions where access to experienced radiologists is limited. Its capability to deliver near-instant diagnostics and natural language explanations also holds promise for use in telemedicine and global public health initiatives.

However, there are limitations that must be acknowledged. The current model is trained and validated on publicly available datasets, which may not fully capture the heterogeneity of real-world populations. Variations in imaging

Protocols, electronic health record formats, and linguistic differences in clinical notes may impact generalizability. Furthermore, reliance on Gemini for LLM integration raises considerations around computational cost and external API dependency, which could affect deployment success in highly constrained environments.

Another limitation is the system's binary classification approach, which simplifies malignancy risk into a yes-or-no prediction. While this is suitable for screening purposes, future iterations should explore probabilistic outputs or risk stratification to guide treatment planning. Additionally, ethical and legal concerns regarding AI-assisted diagnosis, including accountability, data governance, and patient consent, remain critical areas for longer-term adoption.

Despite these limitations, PulmoAID presents a strong, adaptable foundation for AI-assisted diagnosis. Its potential lies not just in improving accuracy, but in making advanced diagnostics accessible, interpretable, and scalable. With further validation and thoughtful refinement, it could become a practical tool for supporting clinicians around the world.

CONCLUSIONS and FUTURE RESEARCH

This study demonstrates the power and practicality of multimodal fusion in advancing lung cancer diagnostics. By combining visual features extracted from CT scans with language-based insights from clinical notes and key patient demographics, the proposed approach outperforms traditional image-only models by a significant margin. Our fusion model, which leverages 2D VGG16 slices alongside clinical predictors in a logistic regression framework, achieved an F1 score of 88%, substantially higher than the 62% score from the CT-only model. This result highlights the value of integrating cross-modal data to capture both structural and contextual cues in disease progression.

Interestingly, the 2D transfer learning approach outperformed more complex 3D CNN architectures like 3DResNet and MedicalNet. This can be attributed to the inherent variability and inconsistency in 3D CT volumes across datasets, suggesting that simpler, slice-based models may be more robust and generalizable for certain applications. Furthermore, the LLM-based textual pipeline not only helped boost predictive performance but also enriched interpretability, allowing the system to factor in nuanced patient histories, risk factors, and comorbidities, elements typically overlooked in image-based models alone.

One of the key outputs of this research is PulmoAID, a scalable, secure, and intuitive platform that democratizes access to high-quality lung cancer diagnostics. Its ability to serve as a "digital radiologist" at the point-of-care, particularly in rural clinics, underserved regions, or emergency rooms. It has the potential to reduce diagnostic delays and health disparities. By translating complex clinical data into human-understandable recommendations, PulmoAID paves the way for more equitable and personalized precision medicine.

Looking ahead, the next phase involves clinical validation across multiple centers and regions. This step is essential for understanding how the system performs across diverse patient populations, imaging protocols, and clinical contexts. Future iterations will also move beyond binary classification, offering probabilistic risk scores or stratified outputs to better inform downstream decision-making.

We also plan to expand the system's scope by integrating genomic and pathology data and exploring visual-language models that can jointly reason across images and text. These advancements would bring PulmoAID closer to functioning as a comprehensive diagnostic assistant, capable of supporting a wide range of pulmonary diseases—not just lung cancer, but also COPD, tuberculosis, and beyond.

Finally, continued work will focus on seamless integration with electronic health records (EHRs) and telemedicine platforms, ensuring PulmoAID can be used at the point of care. Through these developments, we aim to not only improve diagnostic accuracy, but also enhance trust, transparency, and access in AI-driven healthcare for lung cancer screening.

METHODS

Research Design

We systematically defined the design space for our multimodal diagnostic framework to evaluate the impact of different predictor types, feature extraction strategies, and classification algorithms. We tested both CT-only and multimodal configurations that integrated clinical and demographic variables as well as LLM-derived sentiment features.

For CT feature extraction, we implemented multiple transfer learning backbones, including 3D ResNet ((K. Hara, H. Kataoka, Y. Satoh, Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet? In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 6546–6555 (2018).)) , MedicalNet ((S. Chen, K. Ma, Y. Zheng, Med3D: Transfer learning for 3D medical image analysis. arXiv preprint arXiv:1904.00625 (2019). doi:10.48550/arXiv.1904.00625)), and 2D VGG16 to capture diverse spatial and contextual representations. We trained a variety of machine learning classifiers, including logistic regression, XGBoost, random forest, k-nearest neighbors, and naïve Bayes, each with specific hyperparameter grids. As part of building an accurate and robust system, parametric studies on relevant training parameters were performed to arrive at the best models in terms of minimizing false positives and false negatives. We optimized hyperparameters through exhaustive or randomized search over learning rates, solvers, numbers of estimators, and smoothing parameters for traditional ML models, as well as network depth, optimizers, batch size, and epochs for deep learning architectures. This search space as summarized in Table 1 ensured that our final model architecture and parameters produced robust predictions while allowing fair comparisons across configurations.

Independent Variables:

Variable	Levels/Settings
Predictor(s)	[CT only, Multimodal]
CT Feature Extraction	[3D ResNet, MedicalNet, 2D VGG16]
Classifier	[Logistic, XGBoost, Random Forest, KNN, Naïve Bayes]

Hyperparameters:

AI Algorithms	Hyperparameters	Range/Settings
ML		
K Nearest Neigbors (KNN)	n-neighbors	[1-50]
Logistic Regression (LR)	Solvers	[lbfgs, liblinear]
XGBoost (XGB)	Learning Rate	[0.01-1]
Random Forest (RFC)	Number of Estimators	[10-200]
Naïve Bayes (NB)	Smoothing (alpha)	[0.001-10]
DL & Transfer Learning		
3DResNet, MedicalNet	Number of Layers	[1-5]
	Learning Rate	[0.01-1]
VGG16	Optimizer	Adam, SGD
	Batch Size	Various
	Epochs	[1-50]

Table 1: Independent variables and corresponding hyperparameter ranges explored in the design space. The table summarizes the combinations of predictors, CT feature extraction models, and classification algorithms, along with key tunable hyperparameters and ranges used for model selection and optimization.

Data Preparation and Model Training Framework

This section describes the complete end-to-end workflow for our proposed multimodal lung cancer diagnostic system. The framework integrates standardized pre-processing of longitudinal 3D chest CT scans, deep feature extraction via transfer learning with pre-trained 2D and 3D convolutional neural networks, and fusion of imaging features with structured demographic, clinical, and sentiment score as shown in Figure 8. We trained a fully connected neural network for binary classification and applied interpretability techniques to explain model predictions. We followed best practices for data splitting, hyperparameter optimization, and performance evaluation to ensure robustness and clinical relevance of the outcomes.

Imaging Data Acquisition and Preprocessing

In this study, we utilized longitudinal 3D chest computed tomography (CT) scans collected at multiple time points (T0, T1, T2) from publicly available sources such as the National Lung Screening Trial (NLST) Dataset, which included 53,000 participants aged 55 to 74 years who were considered high-risk due to extensive smoking histories. The training subset included 310 subjects with cancer (58042 images) and 369 subjects w/o cancer (50898 images). The Testing subset included 312 subjects with cancer (58982 images) and 183 subjects w/o cancer (30519 images), offering a diverse range of annotated lung scans. Alongside low dose CT (LDCT) images, the dataset includes extensive clinical

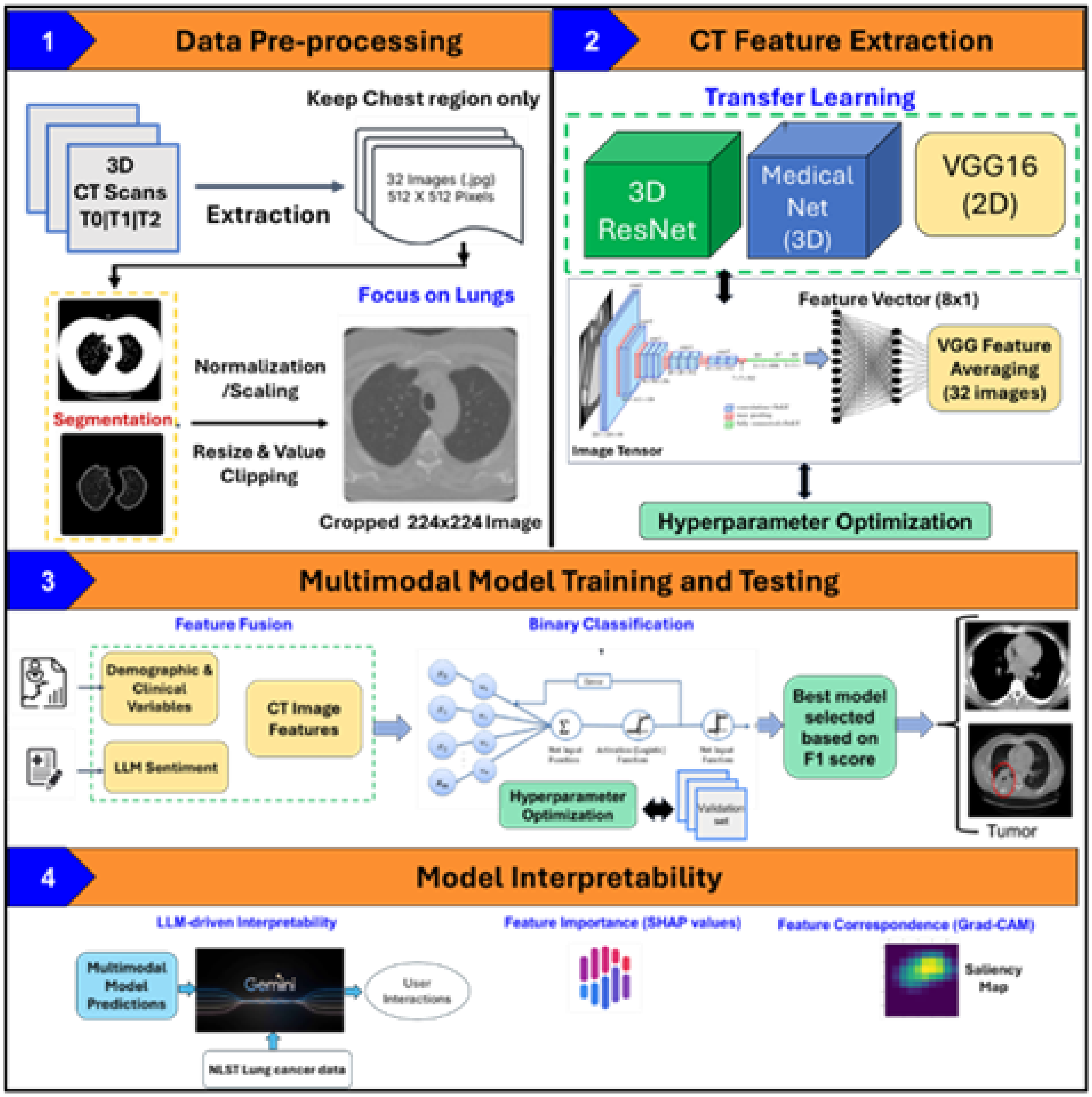

Figure 8: Overview of the multimodal lung cancer diagnostic pipeline. The system consists of four main stages: (1) pre-processing and segmentation of longitudinal 3D chest CT scans, (2) CT feature extraction using pre-trained 2D (VGG16) and 3D (3D ResNet, MedicalNet) convolutional neural networks, (3) multimodal model training combining CT image features with structured demographic, clinical, and sentiment variables, and (4) post-hoc model interpretability using SHAP and Grad-CAM for local and global explanations.

and demographic information.

We processed each CT scan by selecting relevant axial slices, normalizing pixel intensities, and resizing images to match the input dimensions of the pretrained transfer learning models. These steps ensured compatibility across scans obtained using varied acquisition protocols.

For 2D CNNs, we segmented lung regions with a semi-automated approach to remove irrelevant anatomical structures. After segmentation, we resampled each CT volume into 32 representative axial slices per patient, resized each slice to 512×512 pixels in JPEG format, and standardized resolution to maintain uniform model inputs.

Feature Extraction from CT scans

We employed three different algorithms to extract high-level radiomic features from the processed lung CT slices:

1. VGG16 (2D): We fine-tuned an ImageNet-pretrained VGG16 model on labeled lung nodule images. We passed 2D slices through the model, extracted semantic features, and averaged 32-slice feature vectors to generate a fixed-length patient representation. This approach leverages the strong performance of VGG16 on natural images while adapting it for medical imaging contexts.

2. 3D ResNet: This is an extension of the traditional ResNet architecture that uses 3D convolutions instead of 2D, making it well-suited for learning spatiotemporal features in volumetric data of CT scans.

3. MedicalNet: A 3D pre-trained medical imaging model based on 3D ResNet architectures with pre-trained weights optimized for medical image analysis, which accelerates convergence and improves generalizability when training on relatively small labeled medical datasets.

We fine-tuned all models through hyperparameter optimization, including learning rate tuning, batch size selection, and layer freezing strategies. In our studies, 2D CNN models performed better than 3D models due to inconsistencies in slice thickness and availability across scans. We flattened final convolutional outputs and passed them through dense layers to produce high-dimensional feature set linked to spatial patterns.

Structured Clinical Data Integration

We curated structured data features, including patient age, gender, smoking status, comorbidities, and family history of cancer. We encoded categorical variables using one-hot encoding and concatenated them with image features to establish multimodal predictor set.

Clinical Note Processing with LLMs

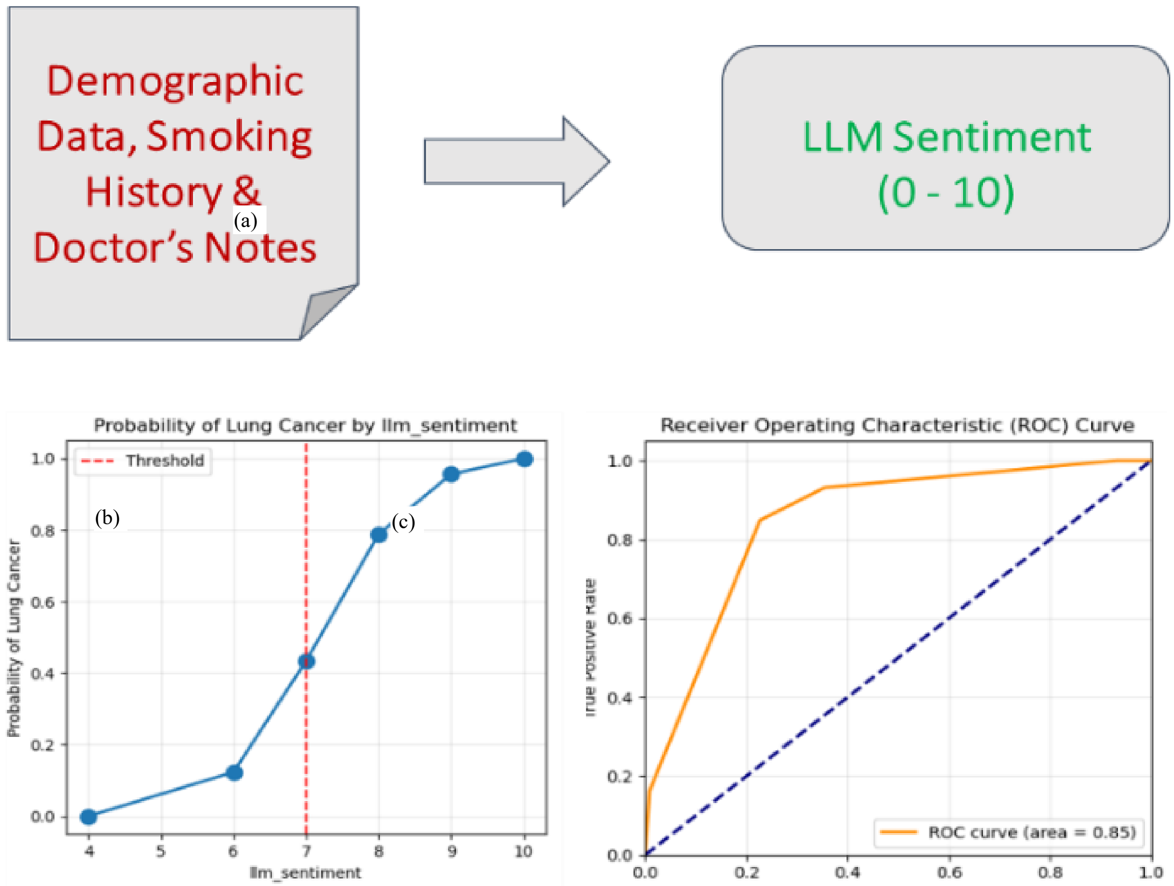

Figure 9: (a) illustration of the LLM-derived sentiment score generated from demographic information, smoking history, and doctor's notes, with score ranging from 0 to 10. (b) Plot depicts the increasing probability of lung cancer as a function of LLM sentiment, with a decision threshold indicated by the red dashed line. (c) Plot presents the ROC curve for the LLM sentiment model, demonstrating a classification performance with an area under the curve (AUC) of 0.85.

We processed unstructured clinical textual data using Gemini framework. After de-identifying, tokenizing, and segmenting notes, we generated semantic embeddings. We used an LLM to produce a cancer-likelihood sentiment score by fine-tuning Gemini with demographic data and physician notes. A score above 7 indicated a strong shift toward high-risk predictions as shown in Figure 9. We vectorized the sentiment score and appended it to each patient predictor set.

Multimodal Model Training and Testing

We concatenated the extracted latent CT features with structured patient data to enable multimodal learning. The final feature set included: (i) latent CT features, (ii) demographic variables (e.g., age, sex, smoking status), and (iii) sentiment feature derived from large language model (LLM) analyzing relevant clinical text.

We designed a fully connected neural network classifier to predict binary outcomes (e.g., tumor or no tumor). We split primary dataset into training and testing sets using stratified sampling and optimized hyperparameters iteratively to maximize performance. We selected the final model based on the F1 score to balance sensitivity and precision in imbalanced classes.

Explainability and Interpretation

To promote clinical trust, we applied Grad-CAM heatmaps to highlight CT regions influencing predictions and computed SHAP values to quantify the contribution of each tabular and LLM-derived feature. We delivered model outputs through a virtual doctor interface that generated natural language explanations using Gemini, summarizing the rationale behind each prediction in clinician-friendly terms.

All the above steps come together in an integrated pipeline implemented as a PulmoAID diagnostic system. We designed the core model architecture of PulmoAID to facilitate effective fusion of features from three distinct modalities: radiological images, structured clinical data, and unstructured text processed by large language models. Based on superior performance, the selected image processing is based on a VGG16 network, truncated before the classification layers to retain spatial and contextual information from CT slices.

Modeling Environment and Libraries

For model development and testing work, Python language was selected for its ease of use and availability of open-source functional libraries.

- Modeling and Computing Environment: VS Code
- Numerical and Statistical Libraries: Gemini, pytorch, torchvision, pillow, pandas, numpy, sklearn, matplotlib, xgboost, shap, seaborn, pydicom, Gemini
- Web Development: Streamlit

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